

Movie Ticket Booking System - Project Report

Course Code: CSC 1031 - Computer Science Assignment

Repository URL: <https://github.com/charan-arunagiri/GroupA-MovieTicketBookingSystem>

Submission Date: July 24, 2026

4.1 Cover Page

- **Project Title:** Movie Ticket Booking System in C
- **Team Name:** Group A[cite: 1]
- **Course Name:** CSC 1031 - Computer Science Assignment
- **Submission Date:** 24th July 2026

Team Members Information

Member Name	Student ID	GitHub Username	Role / Module Assigned
Charan	AS20250385	charan-arunagiri	Team Lead & Core Program Structure
Priyanka	AS20250645	shireepriyanka31-svg	Booking Management (Cancel & Search)
Wageesha	AS20250641	wageeshas123	Revenue & Financial Reporting

4.2 Introduction

For this project, we built a command-line interface (CLI) application in standard C that manages all these theater operations entirely in memory. The system is configured for 3 movies with 2 showtimes each. Each showtime features a 50-seat grid arranged in 5 rows and 10 columns. Seats are divided into Regular, Premium, and VIP sections based on row location, each with different base prices. The application runs through a menu loop that continues running until the user selects the exit option.

4.3 Team Contribution Summary

Team Member	Module Assigned	Work Done & Specific Contribution
Charan	Core Architecture & Booking Engine	Created the main program structure, continuous menu loop, and data structures (<code>structs</code>), [Wrote functions to view showtimes (<code>viewMoviesAndShowtimes</code>), display the interactive 2D seat map (<code>viewSeatMap</code>), handle seat booking (<code>bookATicket</code>), and calculate dynamic prices (<code>calculatePrice</code>)
Priyanka	Booking Management & Search	Wrote the cancellation function (<code>cancelBooking</code>) to free up booked seats and update refund totals. Developed the search function (<code>searchBooking</code>) to find reservations by customer name or seat number, and created a case-insensitive string helper (<code>containsIgnoreCase</code>).
Wageesha	Financial Analytics & Revenue	Implemented the revenue reporting function (<code>showRevenue</code>) to calculate tickets sold per showtime, total earnings per movie, and overall grand total theater earnings.

4.4 Design Overview

1. Data Structures (**structs**)

We structured our data using nested C **structs** to keep track of movies, showtimes, and seats in memory:

- **Seat Struct:** Stores seat status (**isBooked**), customer name, customer category ("Regular", "Student", or "Senior"), seat type ("Regular Tier", "Premium Tier", or "VIP Tier"), and final **pricePaid**.
- **Showtime Struct:** Stores the screening time string and a 2D array of 50 seats (5 rows x 10 columns).
- **Movie Struct:** Stores the movie title and an array of 2 showtimes.

2. Pricing and Discount Calculation Rules

Base ticket prices are assigned based on the seat tier row:

- **Regular Tier (Rows A–B):** Rs. 500
- **Premium Tier (Rows C–D):** Rs. 750
- **VIP Tier (Row E):** Rs. 1000

The system applies discounts according to the following rules:

- **Student Discount:** 10% off the base price.
- **Senior Citizen Discount:** 20% off the base price.
- **Group Discount:** 10% off the base price when booking 4 or more seats in a single transaction.

A customer can qualify for only **one category discount** (Student OR Senior Citizen). However, the **Group Discount can be combined** with either a Student or Senior Citizen discount.

3. Program Flow and Functions

The main function `main()` runs a continuous `while` loop displaying the system menu. The program executes specific functions based on user choices:

- `viewMoviesAndShowtimes()` – Lists all movies and available screening times[cite: 1, 2].
- `viewSeatMap()` – Prints the 5x10 seat grid showing free seats (.) and booked seats (X) with seat tiers
- `bookATicket()` – Takes seat choices, calls `calculatePrice()` to compute the final cost, and saves the reservation
- `cancelBooking()` – Resets booked seats back to free and adjusts the total revenue
- `searchBooking()` – Searches active reservations by customer name or seat coordinate
- `showRevenue()` – Generates a summary of tickets sold and revenue per movie and showtime
- Exit and return to main menu function

4.5 Screenshots & Program Walkthrough

```
=====
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 1
=====

=====
AVAILABLE MOVIES & SHOWTIMES
=====
1. Inception
Showtimes: [10:00 AM] [3:00 PM]
=====
2. Interstellar
Showtimes: [11:00 AM] [6:00 PM]
=====
3. The Dark Knight
Showtimes: [1:00 PM] [8:00 PM]
=====
Press 'Enter' key to go back to the Main Menu
```

1. View Movies and Showtimes

```
=====
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 2
=====
::::::::: Select Movie for Seat Map ::::::::::
1. Inception
2. Interstellar
3. The Dark Knight
Choose Movie (1-3): 2
=====
::::::::: Select Showtime ::::::::::
1. 11:00 AM
2. 6:00 PM
Choose Showtime (1-2): 1
=====
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^ SCREEN ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
          1  2  3  4  5  6  7  8  9 10
[REGULAR] Row A:  [.] [.] [.] [.] [.] [.] [.] [.] [.] [.]
[REGULAR] Row B:  [.] [.] [.] [.] [.] [.] [.] [.] [.] [.]
[PREMIUM] Row C:  [.] [.] [.] [.] [.] [.] [.] [.] [.] [.]
[PREMIUM] Row D:  [.] [.] [.] [.] [.] [.] [.] [.] [.] [.]
[VIP]      Row E:  [.] [.] [.] [.] [.] [.] [.] [.] [.] [.]
=====
LEGEND: [.] = Free | [X] = Booked
=====
```

2. View Seat Map

```

=====
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
-----
Enter your choice (1-7): 3

:::::::::: Select Movie ::::::::::
1. Inception
2. Interstellar
3. The Dark Knight
Choose Movie (1-3): 2

:::::::::: Select Showtime ::::::::::
1. 11:00 AM
2. 6:00 PM
Choose Showtime (1-2): 1

Enter Customer Name: Priyanka

:::::::::: Choose Customer Category ::::::::::
1. Regular
2. Student (10% Discount)
3. Senior Citizen (20% Discount)
Choose Category (1-3): 2

Enter number of seats to book: 2

:::::::::: Seat Selection ::::::::::

```

```

:::::::::: Seat Selection ::::::::::
(10% extra discount applies for Group booking
of 4 or more tickets!)
Enter Seat Code for ticket #1 (e.g. A3): b5
Enter Seat Code for ticket #2 (e.g. A3): b6

=====
BOOKING SUMMARY
=====
Customer Name: Priyanka
Movie: Interstellar
Showtime: 11:00 AM
Category: Student

-----
Seat B5: Rs. 900.00
Seat B6: Rs. 900.00

-----
Total Cost: Rs. 1800.00
=====
Booking successful!

-----
Press 'Enter' key to go back to the Main Menu

```

3. Book a Seat

```

=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 4

::::: Select Movie for Cancellation ::::::
1. Inception
2. Interstellar
3. The Dark Knight
Choose Movie (1-3): 2

:::::::::: Select Showtime ::::::::::::::
1. 11:00 AM
2. 6:00 PM
Choose Showtime (1-2): 1

Enter Seat Code to Cancel (e.g. A3): b5

=====
                        CANCELLATION CONFIRMED
=====
Customer Name: Priyanka
Cancelled Seat: B5
Refund Amount: Rs. 900.00
=====

Press 'Enter' key to go back to the Main Menu

```

4. Cancel Booking


```

=====
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 5

::::::::::::: Search Reservation :::::::::::::::
1. Search by Customer Name
2. Search by Seat Code

Choose Option (1-2): 1

Enter Customer Name (or part of name): priyan

=====
SEARCH RESULTS
=====
Customer: Priyanka      | Movie: Interstellar | Time: 11:00 AM | Seat: B5 | Type: Student | Paid: Rs. 900.00
Customer: Priyanka      | Movie: Interstellar | Time: 11:00 AM | Seat: B6 | Type: Student | Paid: Rs. 900.00
=====

Press 'Enter' key to go back to the Main Menu

```

```

=====
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 5

::::::::::::: Search Reservation :::::::::::::::
1. Search by Customer Name
2. Search by Seat Code

Choose Option (1-2): 2

Enter Seat Code (e.g. A3): b5

=====
SEAT STATUS FOR B5
=====
Movie: Inception      | Time: 10:00 AM | Status: [AVAILABLE]
Movie: Inception      | Time: 3:00 PM  | Status: [AVAILABLE]
Movie: Interstellar    | Time: 11:00 AM | Booked by: Priyanka | Type: Student | Paid: Rs. 900.00
Movie: Interstellar    | Time: 6:00 PM  | Status: [AVAILABLE]
Movie: The Dark Knight | Time: 1:00 PM  | Status: [AVAILABLE]
Movie: The Dark Knight | Time: 8:00 PM  | Status: [AVAILABLE]
=====

Press 'Enter' key to go back to the Main Menu

```

5. Search Booking

```
MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 6
=====
                        FINANCIAL & REVENUE REPORT
=====
Movie: Inception      | Tickets Sold: 0 | Revenue: Rs. 0.00
Movie: Interstellar   | Tickets Sold: 2 | Revenue: Rs. 1800.00
Movie: The Dark Knight | Tickets Sold: 0 | Revenue: Rs. 0.00
=====
                        TIER BREAKDOWN & SUMMARY
=====
Regular Tier (Rows A-B): 2 sold | Revenue: Rs. 1800.00
Premium Tier (Rows C-D): 0 sold | Revenue: Rs. 0.00
VIP Tier      (Row E)   : 0 sold | Revenue: Rs. 0.00
=====
Total Capacity:      300 seats
Total Tickets Sold:   2 seats
Occupancy Rate:       0.67%
=====
TOTAL GROSS REVENUE:   Rs. 1800.00
=====

Press 'Enter' key to go back to the Main Menu
```

6. Revenue Report

```
=====
MOVIE TICKET BOOKING SYSTEM
=====
```

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit

```
-----
Enter your choice (1-7): 7
-----
```

Thank you for using Movie Ticket Booking
System! Goodbye.

Process returned 0 (0x0) execution time : 374.669 s
Press any key to continue.

7. Exit

4.6 Challenges & How They Were Solved

Challenge 1: Program Crashing on Invalid User Input

- **Problem:** Using standard `scanf("%d", &choice)` caused infinite menu loops or program crashes whenever users typed letters or special symbols instead of numbers.
- **Solution:** We resolved this by checking the return value of `scanf`. If invalid input is detected, we flush the standard input buffer using `while (getchar() != '\n');`. This clears invalid characters and safely prompts the user again.

```
=====
                        MOVIE TICKET BOOKING SYSTEM
=====
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel Booking
5. Search Booking
6. Revenue Report
7. Exit
=====
Enter your choice (1-7): 8

Invalid option.
Please select 1-7 from the menu.

=====

Press 'Enter' key to go back to the Main Menu|
```

4.7 GitHub Collaboration Summary

1. Commit History

Commits

main

Commits on Jul 24, 2026

- Update README.md**
charan-arunagiri authored 1 hour ago Verified 275290d
- Update README.md**
charan-arunagiri authored 1 hour ago Verified 1bdc530

Commits on Jul 23, 2026

- Update Group_Project_Submission.txt**
charan-arunagiri authored yesterday Verified 3ca8c8a
- Update README.md**
charan-arunagiri authored yesterday Verified 1c97c37
- feat: add cancelBooking and SearchBooking functions**
shreepryanka11-rsg authored yesterday Verified f0ddcbb
- Add cancelBooking and SearchBooking functions**
shreepryanka11-rsg authored yesterday Verified 816aed3
- Merge pull request #12 from charan-arunagiri/feature/revenue-report**
sagarshas123 authored yesterday Verified a7459eb
- Implement showRevenue function and financial summary report**
sagarshas123 authored yesterday Verified 1955455
- Merge pull request #11 from charan-arunagiri/feature/cancellation-and-search**
shreepryanka11-rsg authored yesterday Verified 549100f
- Add cancelBooking and searchBooking functions**
shreepryanka11-rsg authored yesterday Verified 90dc455
- Merge pull request #10 from charan-arunagiri/feature/ticket-booking-module**
charan-arunagiri authored yesterday Verified c303b4e
- Add bookATicket and calculatePrice functions**
charan-arunagiri authored yesterday Verified 02b304e
- Merge pull request #9 from charan-arunagiri/feature/view-movies-and-seatmap**
charan-arunagiri authored yesterday Verified 2442b48
- Add viewMoviesAndShowtimes and viewSeatMap functions**
charan-arunagiri authored yesterday Verified 6fb824f

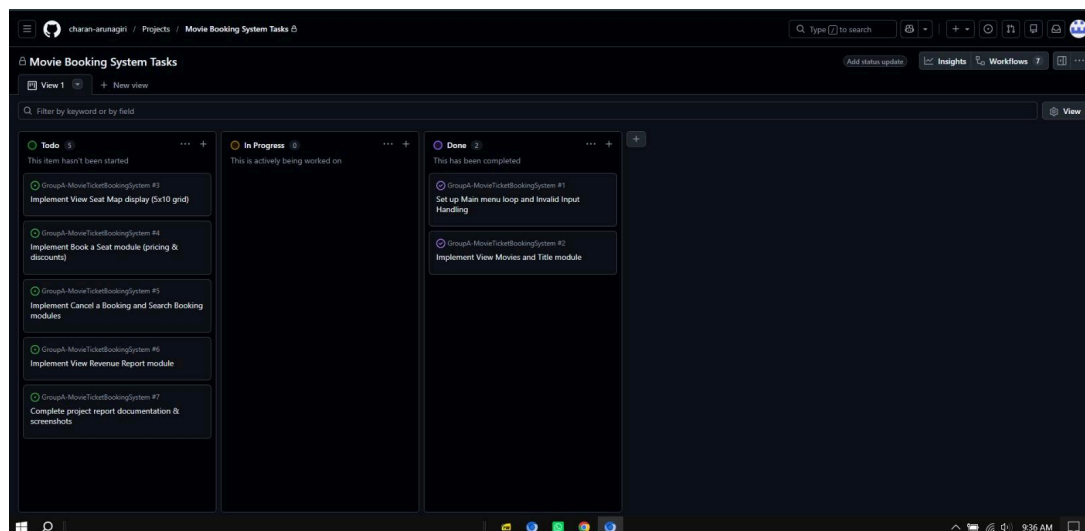
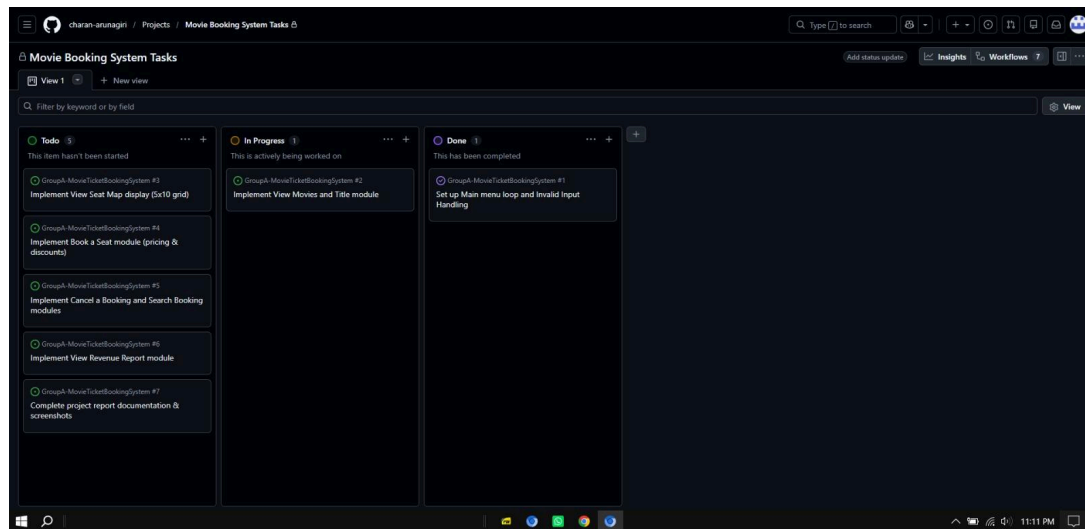
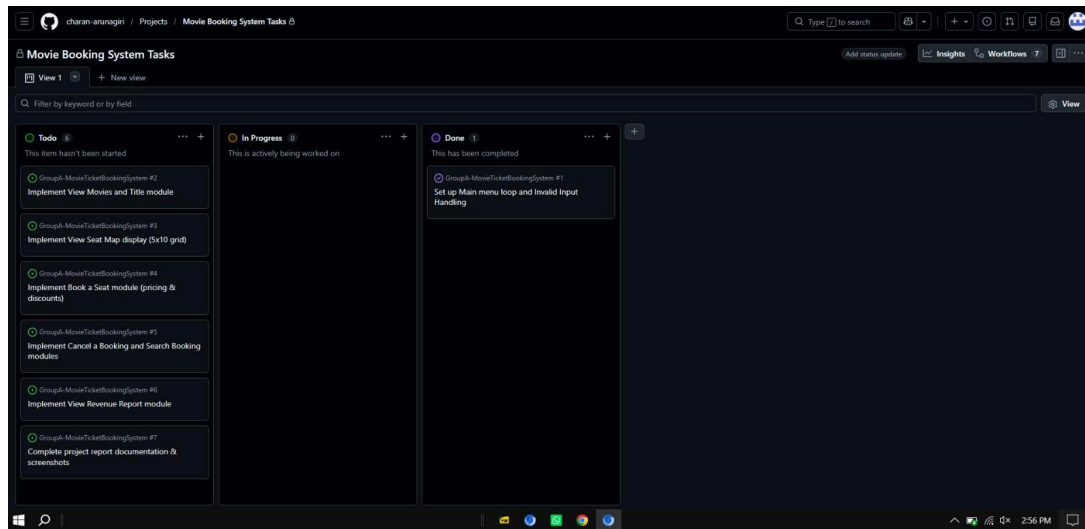
Commits on Jul 22, 2026

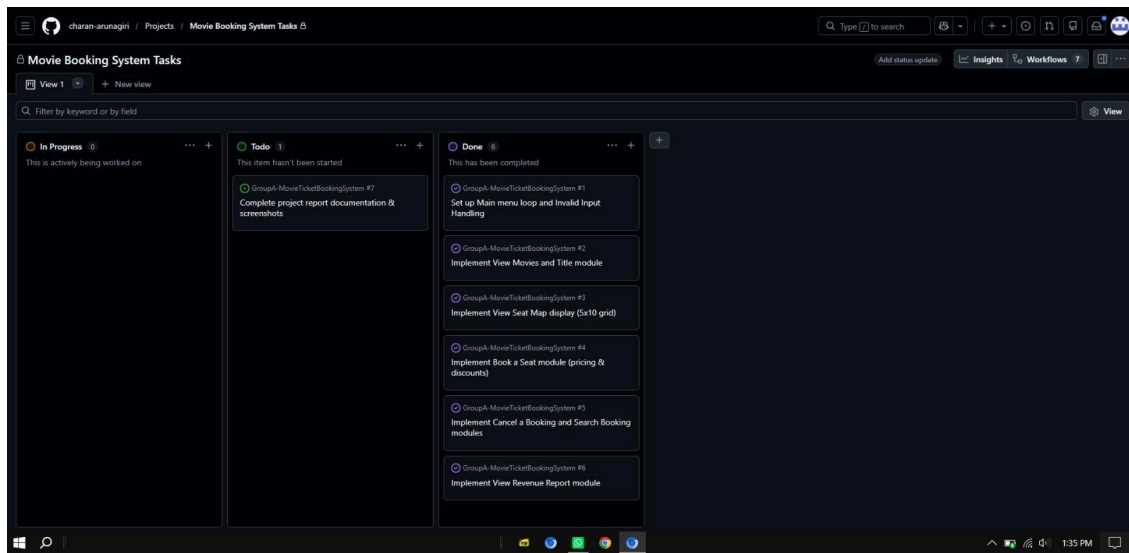
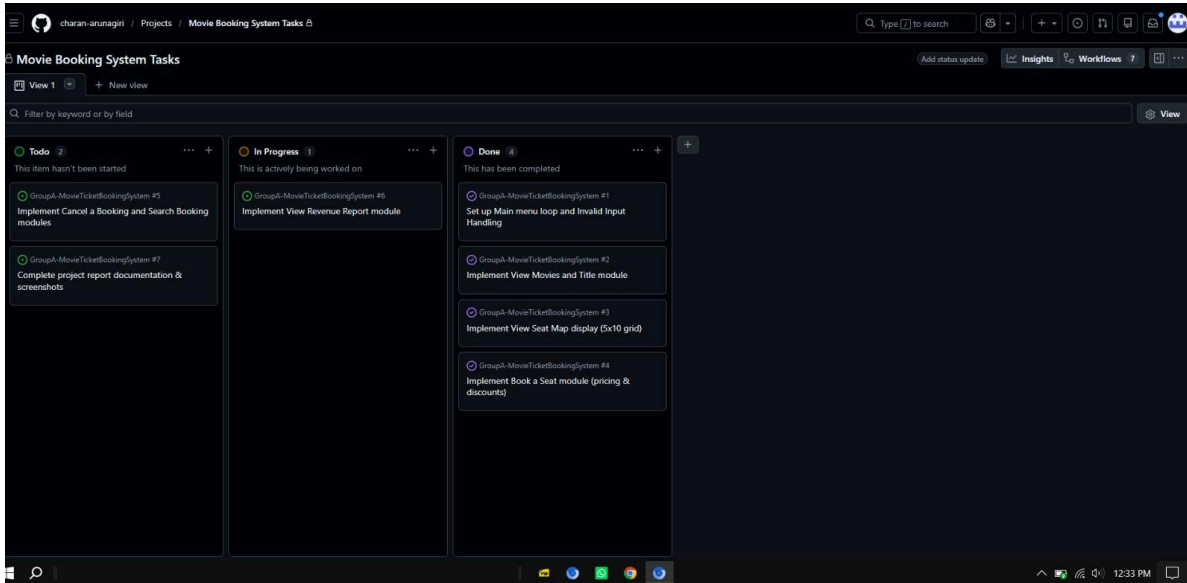
- Merge pull request #8 from charan-arunagiri/feature/system-init-and-menu**
charan-arunagiri authored 2 days ago Verified c4f5bbf
- Add main menu loop, system initialization, and seat parser**
charan-arunagiri authored 2 days ago Verified 83e1623

Commits on Jul 21, 2026

- Create Group_Project_Submission.txt**
charan-arunagiri authored 3 days ago Verified 0ab19e4
- Update README.md**
charan-arunagiri authored 3 days ago Verified 9d5333d
- Initial commit**
charan-arunagiri authored 3 days ago Verified 796a107

2. GitHub Issues & Project Board





3.Reflection on Teamwork

Our team used GitHub as a central workspace to collaborate on the project:

- Charan created the initial repository structure and core main file.
- Priyanka and Wageesha developed their assigned modules in individual branches and pushed their completed features to the repository.
- We tracked tasks through GitHub Issues/Projects, reviewed code changes together, and committed updates frequently to complete the project smoothly before the submission deadline.